

Pranav Ghate

Davis, CA • (213) 421-7271 • pranavghate94@gmail.com • pranavghate.me

Work authorization: U.S. Permanent Resident

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C#, SQL, CSS

Frameworks: Django, Flask, SQLAlchemy, Litestar, Node.js, Vercel, Astro, Tailwind CSS, React, Next.js, Express

Tools: Git, Docker, AWS, VSCode, Visual Studio, Unity3D, S3, Gemini CLI, Gemini Code Assist

Databases: PostgreSQL, MySQL, Microsoft SQL Server, ClickHouse

WORK EXPERIENCE

Software Engineer III - Full Stack

July 2022 - June 2025

University of California, Davis - Plant Simulation Lab

Davis, CA

- Deployed a containerized and scalable data pipeline in Docker to manage complex sensor data, eliminating manual data handling and reducing data access time for a team of 30 researchers.
- Resolved critical data fragmentation by building a high-performance data framework using PostgreSQL, AWS S3 and MinIO, creating a centralized solution to support agricultural research efforts.
- Automated data collection for farming and breeding partners by deploying a software solution that replaced manual Excel spreadsheets, eliminating the need for internal data cleaning and improving data consistency.
- Developed a unified React and Next.js front-end that consolidated multiple data sources into a single, searchable interface.
- Improved developer efficiency by creating a REST API that reduced boilerplate code and accelerated development cycles for new internal tools.
- Programmed native Python APIs and CLI tools to streamline AI workflows, automating pipeline management and providing researchers with direct command-line access to cloud data for model training.

Skills: Python, TypeScript, PostgreSQL, MinIO, AWS S3, Litestar, Flask, React, Next.js, REST APIs

Software Engineer II

July 2018 - June 2022

USC Institute For Creative Technologies - Mixed Reality Labs

Los Angeles, CA

- Led the development of a multi-platform AR exhibit using ARKit, ARCore and Vuforia for mobile devices that engaged 300 participants to educate museum visitors about the Ice Age.
- Partnered with paleontologists and learning scientists on exhibition design and directly contributed to research that demonstrated a 30% increase in learning retention and 90% user satisfaction among museum visitors.
- Contributed key data on the viability of spatial audio for navigation by developing and testing a proof-of-concept Android application, informing the lab's future research strategy in this domain.
- Implemented visual abstraction algorithms on raw drone data to reduce terrain file size and rendering times by 85% without affecting research task performance.

Skills: C#, Unity3D, ARKit, ARCore, Vuforia, Mixed Reality, HCI

Software Engineering Intern

Feb 2017 - May 2018

USC Institute For Creative Technologies

Los Angeles, CA

- Automated the capture of public speaking evaluation data by building a MySQL metadata store from the ground up, which enabled large-scale media indexing and retrieval for research.
- Developed a responsive web interface in React that used Watson API to provide real-time feedback on speech delivery for participants and evaluators, leading to a 40% improvement in evaluation scores.
- Reduced evaluator fatigue by developing a unified dashboard that integrated voice transcription and body language analysis, allowing for a comprehensive, single-view evaluation of participant performance.

Skills: JavaScript, C#, MySQL, React, Express, Webpack, REST APIs, ActiveMQ, Unity3D

EDUCATION

University of Southern California

Los Angeles, CA

Master of Science in Computer Science

August 2016 - May 2018

University of Mumbai

Mumbai, India

Bachelor of Engineering in Information Technology

June 2012 - May 2016